

Causal Inference: Foundations & Applications

Chapter 1: A Definition of Causal Effect

Applied Use Case: Did the Retention Offer Actually Reduce Churn?

1. Problem Context & Variable Notation

To evaluate whether a marketing campaign or customer retention initiative is effective, we define the key random variables within a formal causal framework:

- **Treatment / Intervention (A):**
 - $A = 1$: Customer received a retention offer (e.g., discount, free upgrade, or targeted incentive).
 - $A = 0$: Customer did not receive a retention offer.
- **Outcome (Y):**
 - $Y = 1$: Customer cancelled their subscription (*Churn*).
 - $Y = 0$: Customer maintained their subscription (*Retention*).

2. Individual Causal Effects & Counterfactual Outcomes

For every individual customer i , there exist two **potential (counterfactual) outcomes**:

- $Y_i^{a=1}$: The churn outcome for customer i if they **received** the retention offer.
- $Y_i^{a=0}$: The churn outcome for customer i if they **did not receive** the retention offer.

An **individual causal effect** exists for customer i if and only if:

$$Y_i^{a=1} \neq Y_i^{a=0}$$

Four Customer Response Types (Latent Subpopulations)

Customers can be categorized into four mutually exclusive response types based on their joint potential outcomes ($Y^{a=0}, Y^{a=1}$):

1. **Helped (Saved / Responsive)** ($Y^{a=1} = 0, Y^{a=0} = 1$):
Without the offer, the customer churns ($Y = 1$); with the offer, the customer stays ($Y = 0$). The intervention produces a **beneficial causal effect**.
2. **Immune (Loyal)** ($Y^{a=1} = 0, Y^{a=0} = 0$):
The customer stays regardless of whether they receive the offer. The intervention has **no causal effect** (the discount is a redundant cost).
3. **Hurt (Sleeping Dogs / Triggered)** ($Y^{a=1} = 1, Y^{a=0} = 0$):
The customer would have stayed without the offer, but receiving the offer prompts cancellation (e.g., reminding them of an unused subscription). The intervention produces a **harmful causal effect**.
4. **Doomed (Lost Cause)** ($Y^{a=1} = 1, Y^{a=0} = 1$):
The customer churns regardless of whether they receive the offer. The intervention has **no causal effect** (marketing spend is wasted).

3. Core Principles & Identification Assumptions

Fundamental Problem of Causal Inference

For any single customer i , we only observe the outcome corresponding to the treatment actually received ($Y = Y^A$). The counterfactual outcome under the alternative treatment remains unobserved (missing data problem).

Consistency

$$Y = Y^A = AY^{a=1} + (1 - A)Y^{a=0}$$

The observed outcome Y under actual treatment status A equals the potential outcome under that specific treatment level.

Interference / SUTVA (Stable Unit Treatment Value Assumption)

- **No Interference:** One customer receiving a retention offer does not affect another customer's decision to churn (e.g., no public discussion forums or peer-referral spillovers).
- **Treatment Variation Irrelevance:** The retention offer is well-defined and uniform across all subjects (e.g., exactly 20% discount for 3 months), rather than varying arbitrarily by customer service agent.

4. Average Causal Effects in the Population

Because individual counterfactual outcomes are unobservable, causal inference focuses on the **Average Causal Effect (ACE)** across a population:

$$\Pr[Y^{a=1} = 1] \neq \Pr[Y^{a=0} = 1] \quad \text{or equivalently} \quad E[Y^{a=1}] \neq E[Y^{a=0}]$$

Causal Effect Measures (Example with 100,000 Subscribers)

Suppose in a subscriber population:

- If **everyone** received the offer, churn would be 15% ($\Pr[Y^{a=1} = 1] = 0.15$).

- If **nobody** received the offer, churn would be 20% ($\Pr[Y^{a=0} = 1] = 0.20$).

1. Causal Risk Difference (RD):

$$RD = \Pr[Y^{a=1} = 1] - \Pr[Y^{a=0} = 1] = 0.15 - 0.20 = -0.05 \quad (-5 \text{ percentage points})$$

2. Causal Risk Ratio (RR):

$$RR = \frac{\Pr[Y^{a=1} = 1]}{\Pr[Y^{a=0} = 1]} = \frac{0.15}{0.20} = 0.75 \quad (25\% \text{ relative risk reduction})$$

3. Number Needed to Treat (NNT):

$$NNT = \frac{1}{|RD|} = \frac{1}{0.05} = 20$$

Business Interpretation: The company must send retention offers to **20 customers** to prevent **1 customer** from churning.

5. Causation vs. Association

A critical distinction in Data Science and executive decision-making:

Association (Observational / CRM Data):

$$\Pr[Y = 1 \mid A = 1] \quad \text{vs.} \quad \Pr[Y = 1 \mid A = 0]$$

Compares observed churn rates between two **different subsets** of the population (those who happened to receive the offer vs. those who did not).

Causation (Potential Outcomes):

$$\Pr[Y^{a=1} = 1] \quad \text{vs.} \quad \Pr[Y^{a=0} = 1]$$

Compares churn rates in the **entire population** across two hypothetical counterfactual states.

Why Association $\neq$ Causation in CRM Data

In CRM systems, retention offers are targeted at customers with **high baseline churn risk** ($L = 1$).

- Group $A = 1$ consists predominantly of dissatisfied customers (baseline risk = 40%).
- Group $A = 0$ consists predominantly of satisfied customers (baseline risk = 5%).

Even if the offer reduces churn among targeted high-risk customers from 40% to 25%, a naive observational comparison shows:

$$\Pr[Y = 1 \mid A = 1] = 25\% \quad > \quad \Pr[Y = 1 \mid A = 0] = 5\%$$

A naive associational analysis erroneously concludes that the retention offer *increases* churn. This discrepancy is caused by **confounding bias**.